

Gauss's Law

Prerequisite

- Electric Charge
- Coulomb's Law
- Electric Field
- Electric Flux

Electric Charge

Definition

Electric charge is a fundamental property of matter that gives rise to electric forces. There are two types of electric charge: positive (+) and negative (-). Like charges repel each other, and opposite charges attract.

Units

The SI unit of electric charge is the Coulomb (C). The elementary charge, represented by the symbol e , is the charge of a proton or an electron, approximately 1.602×10^{-19} C

Quantization

Electric charge is quantized, meaning it exists in discrete units. The elementary charge is the smallest possible unit of electric charge.

Conservation of Charge

The total electric charge in an isolated system remains constant. Charge cannot be created or destroyed; it can only be transferred between objects.

Conductors and Insulators

- **Conductors:** Materials in which electric charges can move freely. Metals are good conductors.
- **Insulators:** Materials that do not allow the easy movement of electric charges. Rubber and glass are examples.

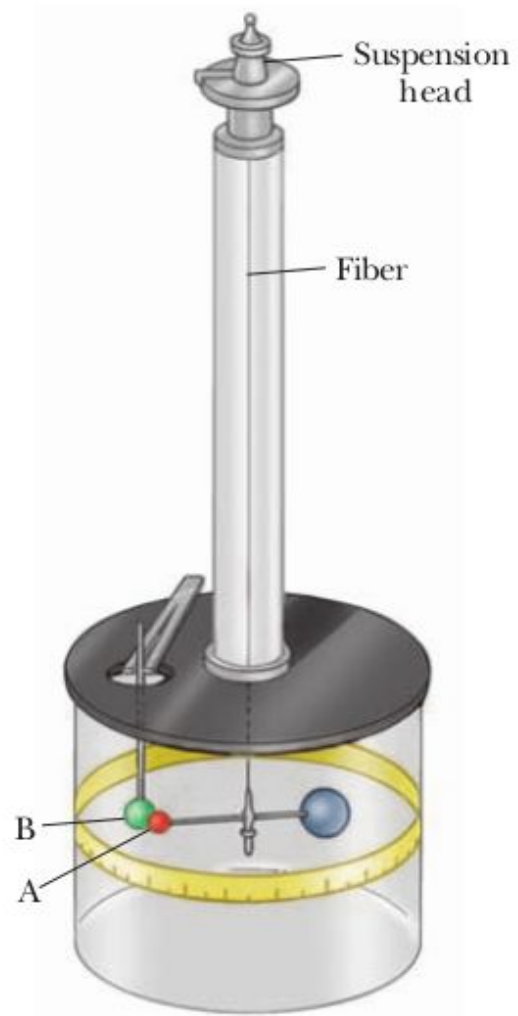
Charging Methods

- Frictional Charging: Charging by rubbing two objects together.
- Conduction: Charging by direct contact between charged and uncharged objects.
- Induction: Charging without direct contact by bringing a charged object near an uncharged object.

Coulomb's Law

Point charge

A point charge is a theoretical concept in physics that represents a particle with all its electric charge concentrated at a single point in space. In reality, particles like electrons and protons have finite sizes, but for certain calculations and theoretical considerations, treating them as point charges simplifies the analysis.



Coulomb's Law

$$F_e = k_e \frac{|q_1||q_2|}{r^2}$$

F_e = Force

k_e = Coulomb's Constant ($8.987 * 10^9 \text{ Nm}^2/\text{C}^2$)

$q_1 q_2$ is charge of particle

r^2 is the distance between particles

Coulomb's constant is also written as $1/4\pi\epsilon_0$

where the constant ϵ_0 (Greek letter epsilon) is known as the permittivity of free space and has the value

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N m}^2)$$

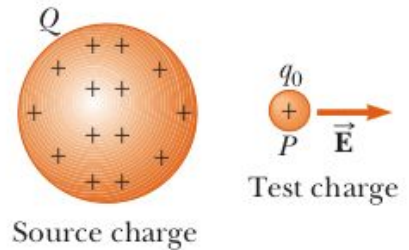
Practice Problem

The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately 5.3×10^{-11} m. Find the magnitudes of the electric force

Electric Field

- The electric field at a point in space represents the force experienced by a positive test charge placed at that point.
- It is a vector field, meaning it has both magnitude and direction.

In this approach, an electric field is said to exist in the region of space around a charged object, the source charge. When another charged object—the test charge—enters this electric field, an electric force acts on it.



Electric Flux

- Electric flux is a measure of how much of an electric field penetrates a given surface.
- It quantifies the number of electric field lines passing through a surface.

Mathematically

Electric Flux Through a Surface (ΦE):

- For a uniform electric field ($\mathbf{E}$) and a surface with area (A):

$$\Phi E = \mathbf{E} \cdot \mathbf{A}$$

ΦE : Electric flux.

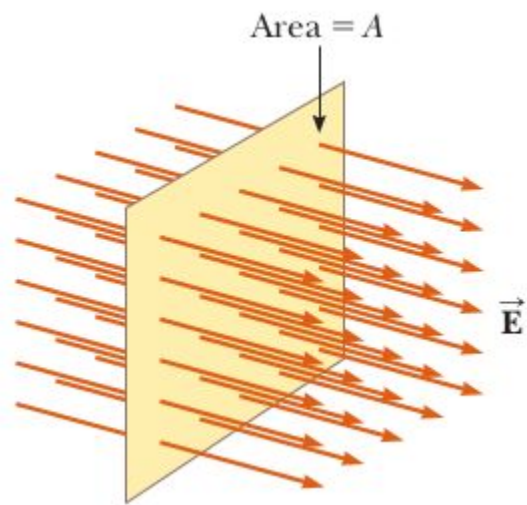
$\mathbf{E}$: Electric field vector.

$\mathbf{A}$: Area vector perpendicular to the surface.

General Case

$$\Phi E = \int \mathbf{E} \cdot d\mathbf{A}$$

- Electric flux through a surface is the surface integral of the electric field over the surface.



Characteristics

Direction:

- Electric flux is positive when the electric field and the normal to the surface point in the same direction.
- Electric flux is negative when the electric field and the normal to the surface point in opposite directions.

Units

SI Unit:

- Volt-meters ($\text{V}\cdot\text{m}$).

Gauss's Law

Gauss's Law is a fundamental principle in physics that relates the electric flux through a closed surface to the total electric charge enclosed by that surface. It is a key concept in the study of electrostatics and is named after the German mathematician and physicist Carl Friedrich Gauss. Gauss's Law is a mathematical representation of one of Maxwell's equations, which describe the behavior of electric and magnetic fields.

Mathematical Formulation

The mathematical form of Gauss's Law for electricity is given by:

$$\oint E \cdot dA = Q_{enc} / \epsilon_0$$

In words, the flux ($\oint E \cdot dA$) of the electric field (E) through a closed surface is equal to the total charge (Q_{enc}) enclosed by that surface divided by the permittivity of free space (ϵ_0).